

Publishing Quantitative and Computational Research

Introductions

1. Welcome and aim of the workshop
2. Who we are, workshop structure
3. Quick participant round: name, field, topic



Who we are



Hour 1

Publishing – not just one thing...

Introduction
Guest: Alie Lassche

- The humanities computing publishing field
- & How to navigate as a quantitative humanities scholar

Hour 2

From idea to paper

Presentation
Exercise on case
Exercise on reviews

- Operationalizing your ideas clearly
- Neuralgic points of paper clarity/structure

Hour 3

Venues & strategies

Presentation
Exercise on venues searching

- Some directions in the field
- Matching vs framing

Hour 4

Transparency, questions, conclusions

- On additional work beyond writing & matching
- Conclusions
- Q&A

10.00	Intro
10.15	Guest: Alie Lassche
10.35	Q&A
10.45	Break (15 min)
11.00	Presentation paper making
11.20	Exercise on paper making
11.50	Exercise on reviewing
12.10	Break (20 min)
12.30	Presentation venues
12.40	Exercise on venues
13.25	Break (5 min)
13.30	Presentations Transparency & open science
13.40	Conclusions, Q&A

Guest: Alie Lassche



Alie Lassche: navigating the publishing field as a computational humanities scholar

From qualitative to quantitative researcher

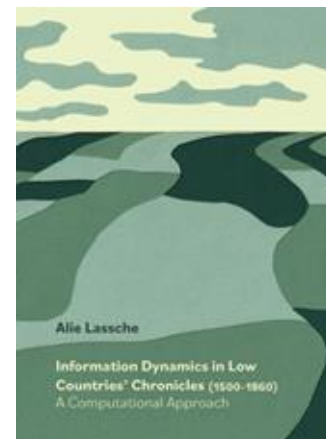
- BA Dutch Language and Culture (2017, Utrecht University)
- RMA Dutch Literature and Culture (2019, Utrecht University) → first introduction to DH/CH
- PhD Dutch History (2024, Leiden University)
- Postdoctoral researcher (Center for Humanities Computing, Aarhus University)

Interests

- Using computational, quantitative and statistical methods to trace the development of literary culture and the dissemination of information in early modern Europe
- Particular focus on communities that were historically marginalized or considered peripheral, whether for political, social, religious, or geographical reasons

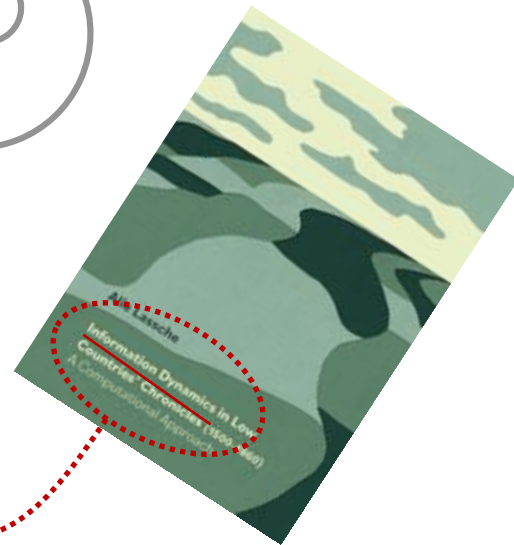
Three (past and ongoing) challenges

- Dissertation: monograph or article-based?
- Where to publish, and how to adapt your writing?
 - Early Modern Low Countries, Quaerendo, PLoS ONE, Nature Reviews Methods Primers, Urban History
 - CHR, LREC, LaTeCH-CLfL, NLP4DH, ACL
- How to explain an interdisciplinary publication list on your CV?
 - Publication types: conference proceedings, journal articles, book chapters, monographs





Q & A



BREAK



Hour 2: From idea to paper



From Vague Idea to Publishable Argument

A structured journey from curiosity to clarity



Good questions start the journey. Good design carries it forward. Clear arguments light the way.

Essential Quantitative/Computational Structure

1. Is there a basic logic to a publishable paper?
2. The Question (ideal center)
3. Methods and results as the core
4. Very important for us/humanists: the relation of pipeline and argument
5. From curiosity to measurable research and research design

Question: Are forests and gardens associated with different emotional/narrative functions in fairy tales?

Method: Computational analysis of lexical occurrence and sentiment patterns in a corpus of fairy tales.

Result: Forests cluster with fear and uncertainty, gardens cluster with safety and order.

Essential Quantitative/Computational Structure

Computational work as structured evidence seeking

1. A researcher suspects that vampire novels become more domestic over time.
1. They use computational methods to search a **large corpus** for changes in **semantic fields** related to home, family, religion, and danger.
1. The algorithm is helping the researcher seek **patterned evidence** in a structured way.

Operationalization: Turning Concepts into Measures

How a humanistic idea becomes structured evidence



Operationalization gives a concept a form that evidence can test.

Making an answerable hypothesis

1. Topic and research question are very different
 - a. Gender in Victorian fiction
 - b. Do female and male characters in Victorian fiction receive different evaluative descriptions?
2. The hypothesis is your question turned into a prediction
 - a. Do female and male characters in Victorian fiction receive different evaluative descriptions?
 - b. Female characters in Victorian fiction will link to appearance and emotion, male characters will link to action and authority



- a. We have a corpus of literature, is there something interesting inside?
- b. We expect specific emotional patterns to emerge in the final chapters of 19th-century Gothic novels

Making an answerable hypothesis

1. Linking question to available material
2. Linking question to method
3. Examples of good and weak formulations

Can you tell
the
differences?

- *We expect romantic novels to use a different kind of language from other novels, making them marked and recognizable in style.*
- *Topic modeling will reveal important hidden patterns in a large corpus; contextual embedding-enriched topic models will reveal more hidden patterns.*
- *Romantic novels in the dataset will contain a higher relative frequency of intimacy-related terms than historical novels published in the same decades.*
- *Newspaper reviews published after 1900 will contain more evaluative adjectives associated with originality than reviews published before 1900.*

The Basic Logic of a Publishable Paper

A clear path from research question to argument



Question

What do you want to know?



Dataset

What material will answer it?



Method

How will you test it?



Result

What did you find?



Claim

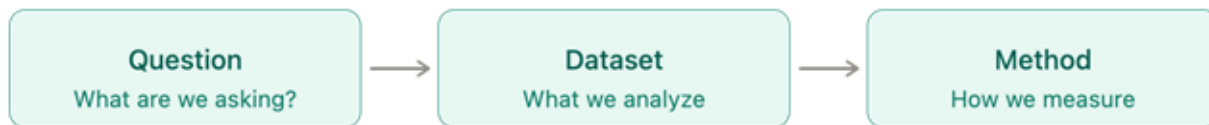
What does it mean?



— Good reasoning connects each step. Good writing makes it clear.

Pipeline: question, dataset, method

1. Start from the question
2. Corpus construction and selection
3. Design the method



Research Pipeline: Question → Dataset → Method

1. QUESTION

Do vampires become more domestic over time?



2. DATASET

100 Vampire Novels (1800~2020)



3. METHOD

Semantic & Sentiment Analysis



and its selected products form the candidate data for the second task, while the entire script and all retrieved products are used for the third subtask.

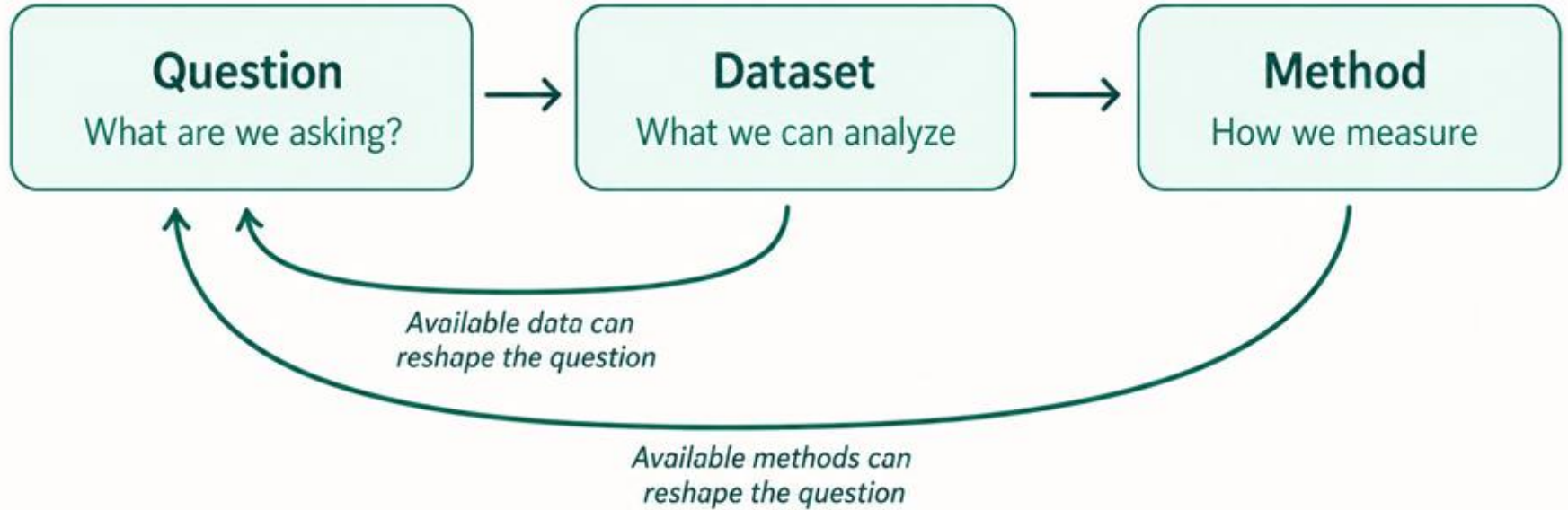
4.4 Human Annotations

Benchmark Annotation: We finally conduct human annotations via Amazon Mechanical Turk (AMT) to provide gold labels for a sampled proportion of data and build them into an evaluation benchmark. For each task, 5,000 data entries are randomly sampled for annotation. We qualify 56 (18.67%) workers from a pool of 300 candidates with excellent annotation records and provide them with detailed instructions to complete each subtask. They are then tasked with annotating (1) the plausibility and feasibility of a given script towards an objective as generated in §4.1, (2) the necessity of purchasing a given product for a specific step in the script, as collected in §4.3, and (3) the overall feasibility of a product-enriched script given the user objective, the entire script, and all retrieved products. Note that only scripts that passed the plausibility annotation are used as candidate data in further tasks. We collect five votes for each entry, and the majority vote is used as the final label. The overall inter-annotator agreement (IAA) is 78% in terms of pairwise agreement, and the Fleiss

5 Experiments and Analyses

5.1 ECOMSCRIPTBENCH Statistics

We first present the statistics of ECOMSCRIPTBENCH in Table 1. In total, we collect 605K scripts with 5.9 million steps. Of a sample of 200 scripts, 94% were annotated as plausibly correct by expert annotators. Among them, approximately 3 million steps are deemed necessary for product purchases by the LLM, while the others do not necessitate products. We also collect 24 million intentions based on 2.4 million products, of which 98.5% from a sample of 200 are deemed plausible by expert annotators, demonstrating the high quality of our dataset. For each task, we collect labels for 5,000 sampled data entries and leave the rest unlabeled. We partition the annotated data into train, dev, and test sets according to an 8:1:1 ratio. Each task follows a high expert acceptance rate, demonstrating the reliability of ECOMSCRIPTBENCH. We further visualize the product distributions of our collected scripts by steps in Figure 3. We observe that as the script progresses (the number of steps increases), more steps are required to associate with products, indicating the need for e-commerce script planning in real-world scenarios.



Creative, but also risky: we may end up asking only what our available data and methods let us see.



Good research uses these feedback loops consciously.

The Architecture of a Paper

Each section has a job in building the argument



— A good paper is not a pile of parts, but a structure in which each section helps carry the argument.

The Architecture of a Paper

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The Architecture of a Paper

1. Title!
2. Abstract: problem, gap, data, method, finding and significance in 3 sentences
3. Introduction
4. Related works
5. Material section: data

Abstract

Graph representation learning has garnered significant attention due to its broad applications in various domains, such as recommendation systems and social network analysis. Despite advancements in graph learning methods, challenges still remain in explainability when graphs are associated with semantic features. In this paper, we present GraphNarrator, the first method designed to generate natural language explanations for Graph Neural Networks. GraphNarrator employs a generative language model that maps input-output pairs to explanations reflecting the model's decision-making process. To address the lack of ground truth explanations to train the model, we propose first generating pseudo-labels that capture the model's decisions from saliency-based explanations, then using Expert Iteration to iteratively train the pseudo-label generator based on training objectives on explanation quality. The high-quality pseudo-labels are finally utilized to train an end-to-end explanation generator model. Extensive experiments are conducted to demonstrate the effectiveness of GraphNarrator in producing faithful, concise, and human-preferred natural language explanations.

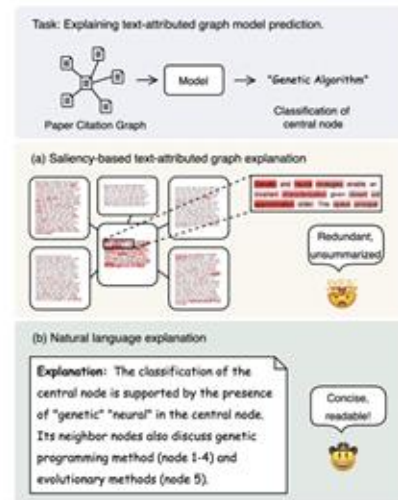


Figure 1: Illustration of saliency-based graph explanation and natural language graph explanation.

recommendation systems, products can be associ-

The Architecture of a Paper

- 1. Method section: how the study works
- 2. Results section

What Figures do you want to include?

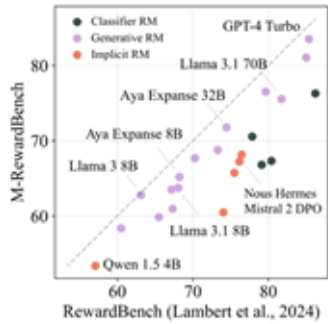


Figure 1: Performance gap between RewardBench (English) and the average M-REWARD BENCH scores across 23 languages for various reward models (Pearson r : 0.92, Spearman ρ : 0.89). All models underperform on our multilingual benchmark compared to their performance on the corresponding English benchmark.

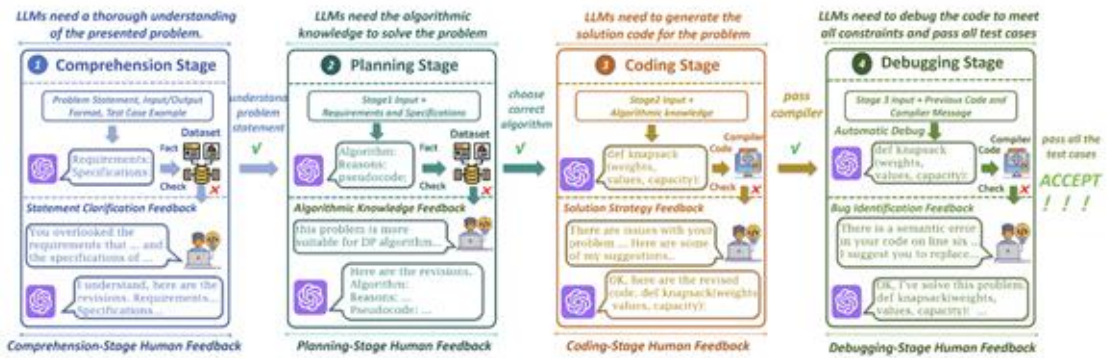
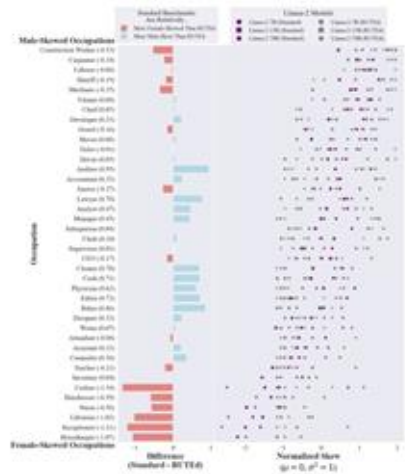


Figure 1: Illustration of ELABORATION evaluation. A human feedback taxonomy, structuring the entire programming process into four stages, enables stage-specific evaluation.



The Architecture of a Paper

1. Method section: how the study works
2. Results section

What Tables do you want to include?

	Neutrality	Skew	Stereotype
Bedtime	-0.07	0.57	0.36
Personas	-0.25	0.54	-0.36
ESL	0.18	-0.39	0.54

Table 1: Rank correlation between standard benchmarks and RUTEd evaluations for each metric.



Prompt: Listwise Ranking Prompt

You are RankLLM, an intelligent assistant that can rank passages based on their relevance to the query.

I will provide you with {num} passages, each indicated by a numerical identifier []. Rank the passages based on their relevance to the search query: {query}.

[1] {passage 1}
[2] {passage 2}
...
[{num}] {passage {num}}

Search Query: {query}.

Rank the {num} passages above based on their relevance to the search query. All the passages should be included and listed using identifiers, in descending order of relevance. The output format should be [] > [], e.g., [4] > [2]. Only respond with the ranking results, do not say any word or explain.

N	Model	DL19	DL20	BEIR Avg
20	RankMistral ₂₀	63.47	59.56	48.45
	RankMistral ₁₀₀	65.82	62.03	49.87
40	RankMistral ₂₀	65.89	63.88	50.07
	RankMistral ₁₀₀	67.77	66.01	51.88
60	RankMistral ₂₀	70.34	64.82	50.66
	RankMistral ₁₀₀	70.97	68.74	52.00
80	RankMistral ₂₀	70.28	68.29	51.13
	RankMistral ₁₀₀	71.36	70.22	52.14
100	RankMistral ₂₀	70.34	69.58	51.85
	RankMistral ₁₀₀	72.55	71.29	52.40

Table 2: Results (NDCG@10) of RankMistral₁₀₀ and RankMistral₂₀ under different passage number *N*.

Selecting Findings

A paper as a story of evidence



1. Cutting secondary or distracting material: balance

Be selective and proportionate—cut what does not matter.



2. Avoiding overclaiming: a classic downfall

Let the evidence support the claim—no more, no less.



A clear, justified claim



3. Don't be afraid of negative or mixed results

They add nuance, prevent bias, and strengthen credibility.



4. Making significance visible

Highlight what matters with appropriate emphasis and clarity.



5. A paper as a story of evidence

Structure the narrative so the reader can follow the logic.



Good papers do not report everything equally; they let the strongest evidence guide the story.

Selecting the Strongest Findings

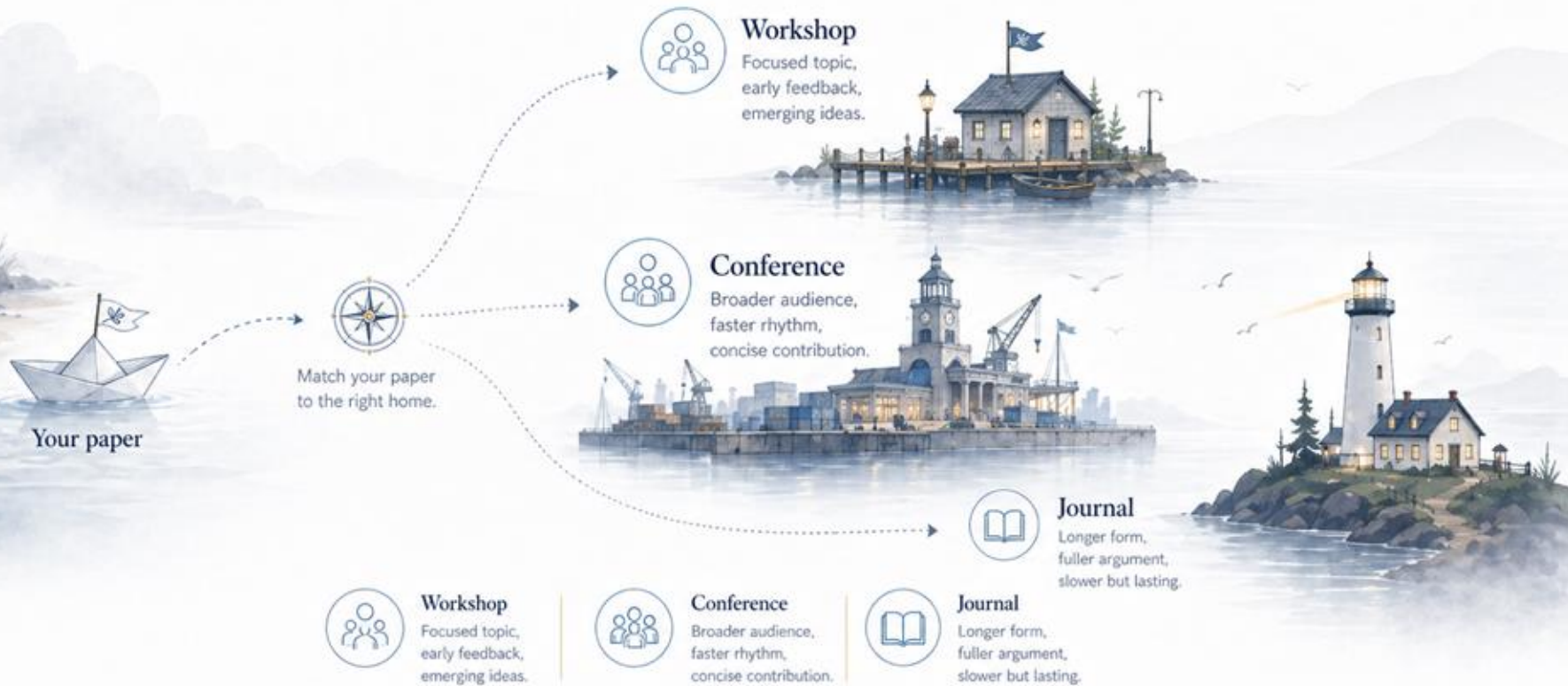
From raw output to the core claim



— A paper becomes stronger when it lets the best evidence carry the claim.

Choosing a Venue

Finding the right harbor for your paper



The best venue is not the grandest one, but the one that fits your argument best.

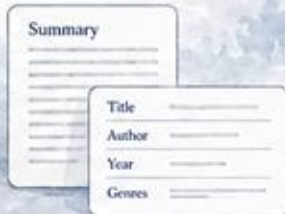
Exercise: From Dataset to Hypothesis

Using the CMU Book Summary Dataset



The dataset

- 16,559 book summaries from Wikipedia
- Metadata such as title, author, publication year, and genres
- Short plot summaries rather than full books
- Rich enough for quick literary and cultural questions



1. Inspect the dataset



2. Choose a topic



3. Write a testable hypothesis



4. Suggest a simple method



Your task (15 min)

- Work in pairs or small groups
- Choose one topic from the next slide
- Formulate 1 hypothesis + 1 simple method
- Be ready to explain what result would support it



A dataset can provoke a question; your job is to make it testable.

Topics You Can Explore

And a very simple example

Possible topics



1. Genre differences



2. Emotion and affect



3. Action and pace



4. Space and world-building



5. Historical period



6. Gender and relationships



Choose one topic and turn it into a testable hypothesis.

Very simple example



Topic:

Genre



Question:

Do science-fiction summaries use more technology-related words than romance summaries?



Hypothesis:

Science-fiction summaries will contain more technology-related vocabulary.



Method:

Build a small lexicon (e.g. *machine*, *robot*, *signal*, *ship*, *laboratory*, *future*) and compare average counts across the two genres.



Support:

Higher average counts in science fiction would support the hypothesis.



Mixed result:

A negative or mixed result is still informative.



Good hypotheses are specific, testable, and tied to what the dataset can actually show.



BREAK



Hour 3: Venues & strategies



What is the humanities computing field like (rn)?

DH/Humanities Computing != one field

Rather, it is a set of partially overlapping publication circuits...

You can say these circuits cluster by “what counts as a contribution”, also since “what counts as evidence” is under negotiation:

- Close readings + light computation work some places...
- Applications w/o use-case work in others...

What counts?

Interpretive contribution with light computation

DH venues often expect an argument first, method second (tool)

Method-driven contribution

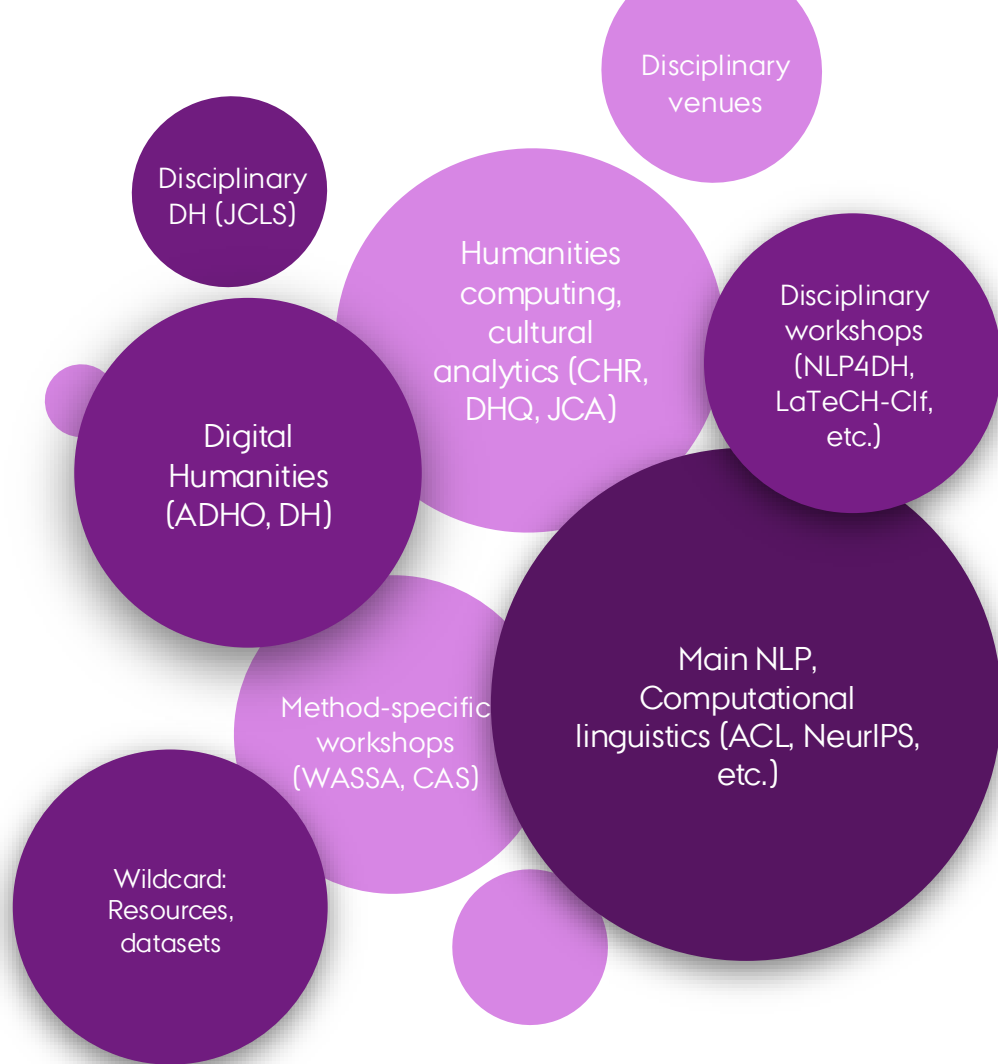
Venues adjacent to NLP care more about what the method does, robustness, scalability, portability

Infrastructure / data / tool building

Less glamorous – very publishable: data papers, tool descriptions, pipelines

Often scattered across venues

HOWEVER they are also stratified by audience!



What audience?

DH-internal venues

(e.g. CHR, Digital Humanities Quarterly)

→ audience already “gets” the methods; you don’t need to justify as much

Disciplinary venues

(literary studies, history, linguistics)

→ the burden flips: you must justify computation and explain method

Technical venues

(e.g. ACL)

→ your (e.g.) literary question matters less than methodological novelty (and harder to impress)

HOWEVER, it also matters what you want to get out of it...

What academic value?

Visibility inside DH (e.g. Digital Humanities Conference, Digital Humanities Quarterly)

- recognizable, legible to the community
- good for positioning yourself within the field

Disciplinary recognition (literary studies, history journals)

- slower, often more skeptical
- may count differently in hiring, promotion, evaluation

Technical credibility (e.g. ACL)

- high bar for methods
- can shift how seriously your work is taken outside the humanities

Output/applications oriented (tools, datasets, tutorials; e.g. Programming Historian, blogs)

- widely used, often highly cited
- but sometimes undervalued in traditional evaluation systems

Selecting a venue

Diagnose your paper/idea

- What is your main contribution? (argument, method, data/tool)
- Who is your primary audience? (DH, disciplinary, technical)
- What do you want to get out of it? (visibility, recognition, credibility, impact)

Read the venue

- Call for Papers (CfP) → what counts as a contribution here?
- Previous publications → what does a “successful paper” look like?

Make the match

- Match paper ↔ venue (venue choice vs framing choice)
- Sometimes a paper doesn't match precisely, and that can be strategic

An incomprehensive list of venues

NLP, methods, technical	EMNLP, ACL, COLING, ICNLSP, NeurIPS, LREC, RANLP
→ Regional	EACL (Europe), NAACL (North America), AACL (Asia), NoDaLiDa (Northern Europe), Clic-It (Italy), SEPLN (Spain), etc.
→ Workshops (DH)	NLP4DH (DH), LaTeCH (humanities, cultural heritage, literature, social sciences), WNU (narrative), LM4DH (DH), etc! (see, e.g., https://aclanthology.org/venues/ws/)
→ Workshops (Methods / Data)	WASSA (sentiment), UnImplicit (language use), CAS (affective science), SocialNLP (SoMe)
DH / Humanities Computing	CHR (Comp. Humanities Research), DH, DSH (Digital Scholarship in the Humanities), DHQ (Digital Humanities Quarterly)
→ Regional	DH Benelux, DHNB, CS2Nordics (Comp. Social Science)
→ Disciplinary / thematic (DH as method)	JCLS (computational literary studies), Plotting Poetry Conference, DraCor (Drama), IC2S2 (social sciences)
→ Mixed approach	Poetics, Journal of Cultural Analytics, Plos One, Nature communications
Disciplinary	New Literary History, Critical Inquiry, MLA, etc.
Infrastructure / tools / pedagogy	The Programming Historian, Open Humanities Data, etc.

Round of introductions

- Brief: research interests
- One/two ideas that you'd like publish on

Max. 2 min

NB. Keep track of the person with your number!

**When you're an introvert and
you hear "let's go around the
room and introduce ourselves"**



Exercise: finding a venue

Take an idea of your partner (with your number)...

5 min: Speculate

- Where could this idea fit in? Does it fit or can it be framed?
- What would match this person's: time horizon, goals, etc?

10 min: Search for venue

- Write down each venue you search and accept/discard & why
- If this was your research interest, which one would you go for?

3-5 min each: Present to your partner!



BREAK



Hour 4: Transparency, Questions, Conclusions



I've written a paper, so I'm ready for submission?

Scientific Journal Peer Review Starter Pack

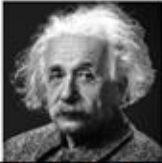
"T-test is not appropriate. Use the Wald-Wolfowitz test with Boole-Bonferroni bound"



"Please perform [series of experiments that would take 6 years]"



Professor the paper was sent to



Masters student who actually reviewed it



"Where are the safety data?"

Right here



	Reviewer #1
	Reviewer #2
	Reviewer #3

"Please acknowledge the following limitations"



But before you do, consider: open science

→ Open Science is many things; one that matters here:

Research transparency → Open data, code, protocols, materials

nature > commentary > article

MENU ▾ **nature**

Commentary | Published: 30 April 1992

The growing inaccessibility of science

Donald P. Hayes

Nature 356, 739–740(1992) | Cite this article

1015 Accesses | 44 Citations | 27 Altmetric | Metrics

Access options

Rent or Buy article	Subscribe to Journal
Get time limited or full article access on ReadCube.	Get full journal access for 1 year
from \$8.99	\$199.00 only \$3.83 per issue
Rent or Buy	Subscribe
All prices are NET prices.	All prices are NET prices. VAT will be added later in the checkout.



Works on my machine...

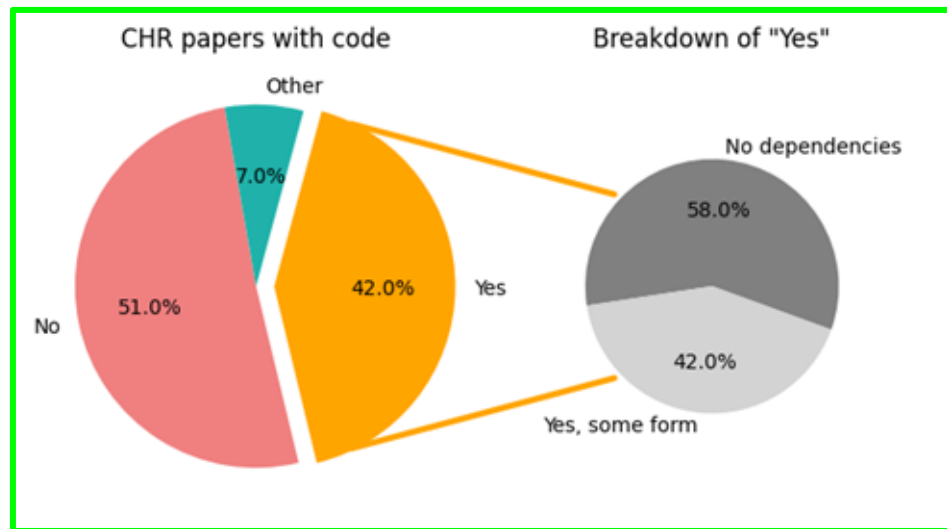
- Code
- Data
- Dependency versioning



Thorsten Ries , Karina van Dalen-Oskam & Fabian Offert

 3612 Accesses  1 Citation [Explore all metrics](#) →

<https://link.springer.com/article/10.1007/s42803-023-00083-w>



<https://anthology.ach.org/volumes/vol0003/works-on-my-machine-case-study-of-replicability-in/>

“Works on My Machine”: A Case Study of Replicability Challenges in Computational Humanities Research

Viktor J. Illmer¹ 

¹ EXC 2020 *Temporal Communities*, Freie Universität Berlin, Germany

Abstract

The replicability of computational research is often stated as a key concern in digital humani-

LLMs: Crisis in reproducibility?

- Changing outputs over time
(e.g. proprietary updates, hidden system prompts)
- No stable versions
(models deprecated, no versioning)
- Underspecified methods
(API use without documentation, unclear pipelines)
- Unspecified prompts
(core part of the method not reported)

DH is now importing a problem that NLP is already struggling with

Good documentation answers not just “what did you do?” but “what could someone else actually reproduce?”

Closed AI Models Make Bad Baselines

🕒 24 minute read

This post was authored by Anna Rogers, with much invaluable help and feedback from Niranjan Balasubramanian, Leon Derczynski, Jesse Dodge, Alexander Koller, Sasha Luccioni, Maarten Sap, Roy Schwartz, Noah A. Smith, Emma Strubell (listed alphabetically)
Header image credit: Sasha Luccioni

<https://hackingsemantics.xyz/2023/closed-baselines/>

Why document & link?

- Technical venues already expect it → reviewers assess whether work is reasonably reproducible

M2. Reproducibility issues	Are there enough details to reproduce the experiments, including hyperparameter selection? Is it clear whether the code/data/etc will be publicly available? If not, is that sufficiently motivated? If so, are they
----------------------------	--

<https://aclrollingreview.org/reviewerguidelines#-task-3-write-a-strong-review>

- Reproducibility increases credibility → others can inspect, test, and build on your work
- Reproducible work travels further → easier to reuse, cite, extend
- It protects your own results → makes errors visible earlier → makes your work interpretable even later (incl. by yourself)

Conclusions

Writing a computational paper

- Focusing on the most salient findings (sifting)
- Making a strong and simply testable hypothesis

Publishing is part of the research

- A paper != a report, but is shaped by venue, audience, and constraints
- Venue choice is not logistical, it frames what counts as a contribution
- Transparency & documentation helps papers become (reusable and verifiable) knowledge

Venue & paper, matching vs framing

- Both are valid strategies





Q & A

